# Keenan Eikenberry

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## Research Interests

- Theory and applications of (self-supervised) representation learning
- Specific interests include transformer architectures, scientific applications, and code representations

## Current Position

2023 – Present
- **Postdoctoral Research Associate**, Dartmouth College
  Department of Mathematics, Hanover, NH
  Supervisors: Yoonsang Lee and Anne Gelb

## Education

2019 – 2023
- **Ph.D. in Applied Mathematics**, Arizona State University
  Advisors: Doug Cochran and Shiwei Lan
  Diss.: *Bayesian Inference for Markov Kernels Valued in Wasserstein Spaces*

2014 – 2016
- **M.A. in Mathematics**, Arizona State University
  Advisors: John Quigg and Steven Kaliszewski
  Thesis: *Functorial Results for $C^*$-algebras of Higher-Rank Graphs*

2009 – 2014
- **B.S. in Mathematics**, Arizona State University
  **B.A. in English Literature (minor in Philosophy)**, Arizona State University
  Summa cum laude, GPA: 4.0

## Papers and Code

### Articles and Preprints

1. **Eikenberry, K.**, Liu, L., & Lee, Y. (2026). Post-training augmentation invariance. *Transactions on Machine Learning Research*. Retrieved from https://openreview.net/forum?id=Z4uUwU6zRe
2. Eikenberry, S. E., Mancuso, M., Iboi, E., Phan, T., **Eikenberry, K.**, Kuang, Y., … Gumel, A. B. (2020). To mask or not to mask: Modeling the potential for face mask use by the general public to curtail the covid-19 pandemic. *Infectious disease modelling*, *5*, 293–308.

### Code

1. Post-Training Augmentation Invariance. (2026).
   https://github.com/keenan-eikenberry/augmentation_invariance.

## Selected Talks

- Wasserstein Dependence and Representation Learning. Applied and Computational Mathematics Seminar, Dartmouth College. November 2024.
- Maximizing Multiview Wasserstein Dependence in Autoencoders. SIAM Mathematics of Data Science Conference, Atlanta, Georgia. October 2024.
- Markov Kernels Valued in Wasserstein Spaces. Online CodEx Seminar. July 2023.

## Selected Talks (continued)

- Stacks of $C^*$-Categories. Great Plains Operator Theory Symposium, Texas A&M University. May 2019.
- Group Actions on Topological k-Graphs. C*-Algebra Seminar, Arizona State University. November 2016.
- $k$-Graphs from Cayley Graphs. Great Plains Operator Theory Symposium, University of Illinois at Urbana-Champaign. May 2016.
- Functoriality for Higher-Rank Graph Algebras. C*-Algebra Seminar, Arizona State University. March 2016.

## Skills

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|---|---|
| Mathematics | Analysis, category theory, probability, optimal transport, partial differential equations, differential geometry, algebraic topology, abstract algebra, operator algebras |
| ML and AI | (Self-supervised) representation learning, neural-guided formal methods, transformer architectures, optimal transport-based methods, theory of deep learning |
| Programming | Python (expert in PyTorch), Julia |